# E.M.I.T. FORENSIC INVESTIGATION REPORT

Case: **Test Case 1: Vulnerable Android APK + Matching Network Evidence** | ID: 044bfbe1-fdbf-41d1-b85a-08f525cd621c | Generated: 2026-09-07 17:09:34 UTC

## 1. Investigation Summary

| | | | |
|---|---|---|---|
| **Case Name** | Test Case 1: Vulnerable Android APK + Matching Network Evidence | **Deterministic Risk** | **HIGH** |
| **Status** | OPEN | **Confidence** | 90% |
| **Evidence Files** | 2 | **Indicators (IOCs)** | 72 |

**Executive Synthesis:** Deterministic Evidence-Weighted Assessment: HIGH Risk (Score: 54, Confidence: 0.9, Evidence Strength: STRONG). Accumulated evidence includes 0 sensitive permissions, 0 suspicious APIs, 33 embedded domains, 6 cross-evidence correlations, and 0 threat intel records.

## 2. Evidence Inventory & Chain of Custody

| File Name | Type | Size | SHA-256 Hash Provenance |
|---|---|---|---|
| DVACv1.apk | APK | 12.32 MB | fbd9ab802d418cc3fbfd89b3cea1c9c4d3cbe20f5f19db79979cbc73ad845516 |
| matching_youtrack.pcapng | PCAPNG | 0.00 MB | 505f6f2ab9b0d2cb44054678a27ca6899ef2511351ee5989f5d64fd322b9fc5c |

## 3. APK Static Findings (OBSERVED EVIDENCE)

| Observation Type | Extracted Value | Confidence |
|---|---|---|
| PACKAGE_NAME | com.zin.dvac | 1.0 |
| DECLARED_PERMISSION | com.zin.dvac.DYNAMIC_RECEIVER_NOT_EXPORTED_PERMISSION | 1.0 |
| DECLARED_PERMISSION | android.permission.USE_BIOMETRIC | 1.0 |
| SUSPICIOUS_PERMISSION | android.permission.READ_CONTACTS | 1.0 |
| DECLARED_PERMISSION | android.permission.READ_EXTERNAL_STORAGE | 1.0 |
| DECLARED_PERMISSION | android.permission.INTERNET | 1.0 |
| DECLARED_PERMISSION | android.permission.WRITE_EXTERNAL_STORAGE | 1.0 |
| EMBEDDED_URL | https://github.com/zinja-coder | 1.0 |
| EMBEDDED_DOMAIN | github.com | 1.0 |
| EMBEDDED_URL | http://schemas.android.com/apk/res-auto | 1.0 |
| EMBEDDED_DOMAIN | schemas.android.com | 1.0 |
| EMBEDDED_URL | http://schemas.android.com/apk/res/android | 1.0 |
| EMBEDDED_DOMAIN | schemas.android.com | 1.0 |
| EMBEDDED_DOMAIN | schemas.android.com | 1.0 |
| EMBEDDED_DOMAIN | github.com | 1.0 |

## 4. Network Findings (OBSERVED EVIDENCE)

| Event Type | Observed Flow / Value | Timestamp |
|---|---|---|
| NETWORK_IP | 127.0.0.1 | 2026-09-07 16:57:48.940000 |
| NETWORK_IP | 127.0.0.11 | 2026-09-07 16:57:48.974000 |
| DNS_QUERY | youtrack.jetbrains.com | 2026-09-07 16:57:49.006000 |

| | | |
|---|---|---|
| DNS_QUERY | youtrack.jetbrains.com | 2026-09-07 16:57:49.018000 |
| DNS_RESPONSE | 63.35.30.167 | 2026-09-07 16:57:49.045000 |
| NETWORK_PROTOCOL | UDP | 2026-09-07 16:57:49.073000 |
| NETWORK_PROTOCOL | DNS | 2026-09-07 16:57:49.104000 |
| NETWORK_PORT | 52200 | 2026-09-07 16:57:49.132000 |
| NETWORK_PORT | 53 | 2026-09-07 16:57:49.159000 |
| NETWORK_PORT | 46230 | 2026-09-07 16:57:49.188000 |
| NETWORK_IP | 127.0.0.11 | 2026-09-07 16:57:49.199000 |
| NETWORK_IP | 127.0.0.1 | 2026-09-07 16:57:49.209000 |
| NETWORK_PROTOCOL | DNS | 2026-09-07 16:57:49.219000 |
| NETWORK_PROTOCOL | UDP | 2026-09-07 16:57:49.231000 |
| NETWORK_PORT | 52200 | 2026-09-07 16:57:49.241000 |

## 5. Deterministic Correlations & Inferred Attack Chain

| Relationship | Class | Strength | Evidence Deduction / Reason |
|---|---|---|---|
| CONTAINS | OBSERVED | STRONG | Application manifest declared permission 'com.zin.dvac.DYNAMIC_RECEIVER_NOT_EXPORTED_PERMISSION'. |
| CONTAINS | OBSERVED | STRONG | Application manifest declared permission 'android.permission.USE_BIOMETRIC'. |
| CONTAINS | OBSERVED | STRONG | Application manifest declared permission 'android.permission.READ_CONTACTS'. |
| CONTAINS | OBSERVED | STRONG | Application manifest declared permission 'android.permission.READ_EXTERNAL_STORAGE'. |
| CONTAINS | OBSERVED | STRONG | Application manifest declared permission 'android.permission.INTERNET'. |
| CONTAINS | OBSERVED | STRONG | Application manifest declared permission 'android.permission.WRITE_EXTERNAL_STORAGE'. |
| EMBEDS | OBSERVED | STRONG | Application bytecode/assets statically embed domain 'github.com'. |
| EMBEDS | OBSERVED | STRONG | Application bytecode/assets statically embed domain 'schemas.android.com'. |
| EMBEDS | OBSERVED | STRONG | Application statically embeds IP address '127.0.0.1'. |
| CONNECTS_TO | OBSERVED | STRONG | Captured network flow establishes active connection to IP '127.0.0.1'. |
| CONNECTS_TO | OBSERVED | STRONG | Captured network flow establishes active connection to IP '127.0.0.11'. |
| OBSERVED_IN | OBSERVED | STRONG | Captured network traffic performed DNS query for domain 'youtrack.jetbrains.com'. |

## 6. External Threat Intelligence (EXTERNAL INTEL)

| Provider | Indicator | Verdict | Score / Details |
|---|---|---|---|
| No threat intel queried. | | | |

## 7. Chronological Timeline (Authentic Dates Only)

| Event | Details | Timestamp |
|---|---|---|
| Network Observation: NETWORK_PORT | Observed Value: 46230 | 2026-09-07 16:57:49.261000 |
| Network Observation: NETWORK_PORT | Observed Value: 53 | 2026-09-07 16:57:49.252000 |
| Network Observation: NETWORK_PORT | Observed Value: 52200 | 2026-09-07 16:57:49.241000 |
| Network Observation: NETWORK_PROTOCOL | Observed Value: UDP | 2026-09-07 16:57:49.231000 |
| Network Observation: NETWORK_PROTOCOL | Observed Value: DNS | 2026-09-07 16:57:49.219000 |
| Network Observation: NETWORK_IP | Observed Value: 127.0.0.1 | 2026-09-07 16:57:49.209000 |
| Network Observation: NETWORK_IP | Observed Value: 127.0.0.11 | 2026-09-07 16:57:49.199000 |
| Network Observation: NETWORK_PORT | Observed Value: 46230 | 2026-09-07 16:57:49.188000 |
| Network Observation: NETWORK_PORT | Observed Value: 53 | 2026-09-07 16:57:49.159000 |
| Network Observation: NETWORK_PORT | Observed Value: 52200 | 2026-09-07 16:57:49.132000 |
| Network Observation: NETWORK_PROTOCOL | Observed Value: DNS | 2026-09-07 16:57:49.104000 |
| Network Observation: NETWORK_PROTOCOL | Observed Value: UDP | 2026-09-07 16:57:49.073000 |

## 8. Technical Uncertainty, Limitations & Analyst Guidance

### *Uncertainty Disclosures:*

• Payload contents were not inspected for TLS-encrypted streams; content cannot be definitively confirmed without session keys.
• Analysis is based on supplied static files and passive network capture without dynamic execution on a physical hardware sandbox.

### *Analyst Guidance & Next Steps:*

> Inspect TLS SNI headers and X.509 certificates to verify remote endpoint ownership.
> Examine decompiled Smali/Java classes around identified suspicious APIs to verify execution control flow.
> Cross-reference observed domains against enterprise proxy logs and external WHOIS registration history.